

Mercury: Facilitating the digital transformation journey for enterprises with Google Cloud

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[About Mercury](#)



To provide its cus with cost-efficient solutions, Mercury leverages the high performance infrastructure of Google Cloud to maintain low maintenance costs and shorten development cycles.

About Mercury

Founded in 2017, Mercury is a technology startup offering cloud-based digital solutions ranging from applications, ERP systems, cloud security to data analytics designed for various industries. Currently providing services to around 100 enterprise customers, Mercury leverages the latest technologies like blockchain, IoT and AI to deliver cost-efficient solutions tailored to customers' needs.

Industries: Technology

RESULTS

- Delivers 2X higher computing performance on Compute Engine
- Simplifies database deployment by automating backups and instantly detecting performance issues with Cloud SQL
- Shortens software development cycles to days on online development environments supported by Cloud Shell
- Helps customers save 80% of maintenance costs through built-

Reduced
application
hosting
costs by
40%

Location: Hong Kong

(<https://cloud.google.com/compute/>)

Google Cloud (<https://cloud.google.com/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud Shell (<https://cloud.google.com/cloud-shell/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine

(<https://cloud.google.com/compute/>)

**in proactive
alert system
in cloud
infrastructure**

As digital transformation becomes a key element in business strategies, more and more enterprises in different industries are moving their legacy IT systems to the cloud for greater scalability and flexibility. However, for companies with limited expertise in cloud technology, the journey of cloud migration can be a long process of trial and error.

This is where Mercury (<https://oss.mtsoln.com/>) steps in. By delivering cloud-based digital solutions tailored to customers' needs in a fast and cost-efficient manner, the Hong Kong-based startup is dedicated to facilitating cloud transformation for enterprises. Founded in 2017, it currently provides a wide range of customized cloud software, including applications, enterprise resource planning systems, cloud security and data analytics, to more than 100 customers in various industries.

"Our mission is to offer the best digital solutions within the shortest possible time by leveraging a hybrid of the most advanced cloud tools, so that our customers can easily enjoy the benefits of cloud migration," says James Huang, lead business architect at Mercury.

At first, Mercury developed and deployed its products on several public cloud platforms, including Google Cloud (<https://cloud.google.com/>). As

time went by, the Mercury team gradually realized that it would be easier to build and manage software for its customers if all its services are running on one single cloud platform, because the talents that are familiar with multiple cloud platforms are scarce. After comparing different offerings, Mercury switched to a single cloud environment on Google Cloud in early 2021 for its high-performance and customizable virtual machines (VMs), as well as its easy-to-use nature.

“Our goal of moving from a multi-cloud to a single cloud environment is to use our resources more efficiently,” explains Huang. “The fact that Google Cloud has an advantage in the performance and flexibility of VMs and provides unparalleled documentation convinced us that it’s the best choice for us to reach the goal.”

Running cost-efficient, high-performance VMs on Compute Engine

Mercury now leverages the VMs on [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) to develop and run all its software services. When Mercury was using other cloud platforms, it sometimes had to buy more central processing units (CPUs) in VMs than it needed, because the CPUs and random-access memory (RAM) in VMs are normally offered in fixed proportions, and Mercury’s products are RAM intensive. Thanks to the VM customization feature of Compute Engine, Mercury is able to purchase computing resources fully according to its needs and save operational costs.

“Compute Engine provides great flexibility for VM configuration, which has helped us and our customers avoid paying for computing resources that are not required,” notes Huang. “With customized VMs, our monthly application hosting costs are reduced by 40%.”

The high performance of the VMs on Compute Engine also allows Mercury to use computing resources more efficiently. According to Huang, the VMs on Compute Engine deliver 2X higher performance than the VMs of the same specification on other platforms. As a result, Mercury successfully helped a logistics company migrate its supply chain system from an infrastructure that required 96 CPUs to a four CPU VM on Compute Engine.

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Simplifying database management with Cloud SQL

Another Google Cloud tool that Mercury has benefited a lot from is Cloud SQL (<https://cloud.google.com/sql>), which it uses to store the data of the cloud software that it developed for its customers. The automated backup feature of Cloud SQL allows Mercury to guarantee full data recovery in case of any data loss, while the well-connected global network of Google Cloud enables the startup to deploy its customers' databases in any location they desire without compromising data transfer speed.

On top of that, Cloud SQL Insights

(<https://cloud.google.com/blog/products/databases/get-ahead-of-database-performance-issues-with-cloud-sql-insights>)

enables Mercury's DevOps engineers to quickly address database performance issues on their own by providing better observability of databases and detecting errors. "Without Cloud SQL, our team would have needed a database administrator to receive error reports from DevOps engineers and resolve database-related issues. It might have taken weeks to tackle a problem. Now, our DevOps engineers are able to clear errors independently and immediately, which has helped increase our work efficiency," notes Huang.

Shortening development cycles with Cloud Shell

Mercury pledges to deliver services to its customers within the shortest possible time and always seeks to accelerate its software development process while ensuring service quality. To that end, its DevOps engineers leverage the web-based development environments supported by Cloud Shell (<https://cloud.google.com/shell>) to write and edit codes directly in the cloud, instead of doing the development work on on-prem devices and uploading the codes to the cloud afterwards. This way, Mercury can not only save time and hardware costs for development, but also prevent errors resulting from moving codes between different environments.

The detailed documentation of Google Cloud also enables the Mercury team to work independently without external help. Huang says that the company's engineers are always able to find the answers they need in the documentation database of Google Cloud, which has helped them upskill themselves and establish best practices for development in a short time.

"Cloud Shell and the easy-to-use nature of Google Cloud have helped greatly shorten the development cycles of our products. With only four full-time staff, we're able to deliver software that is developed and deployed on Google Cloud within one to four days to our customers," he adds.

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Minimizing maintenance effort with automated alerts and Cloud Monitoring

Since Mercury offers software maintenance and monitoring service to all its customers, the team needs a stable cloud infrastructure that requires minimum maintenance effort. Taking advantage of the built-in proactive alert system of Google Cloud, which automatically sends alerts when an error occurs, the Mercury team is able to address operational issues instantly and only has to spend a half day every month to review log details of their products for potential improvements.

“It’s amazing for us and our customers how few resources are required for maintenance work on Google Cloud. We have helped an online marketplace company save 80% of maintenance costs after moving to Google Cloud while ensuring

top software performance at all times,” notes Huang.

For the software products that entail more detailed monitoring, Mercury employs Cloud Monitoring (<https://cloud.google.com/monitoring>) to track operation status, and detect code changes and cyberattacks. This allows engineers to fix operational issues immediately and guarantee operation stability when the software undergoes major changes. For example, when the US government imposed new trade restrictions, Mercury was able to help a logistics company smoothly change the settings of its supply chain system to rearrange shippings.

Continuing optimizing digital solutions with data analytics

Mercury always strives to optimize the cloud solutions designed for its customers. To gain useful insights for software improvements, It plans to leverage Cloud Bigtable (<https://cloud.google.com/bigtable>) and Dataprep by Trifacta (<https://cloud.google.com/dataprep>) to prepare operation history data for in-depth analyses. The startup is also working on becoming a Google Cloud partner, which helps the team understand the underlying technologies of Google Cloud even more deeply and identify the most ideal tools to create better solutions for its customers.

Huang says, “We believe that cloud computing can bring great benefits to every company, and Google Cloud has proven to be a frontrunner in the domain by continuing advancing its capabilities. With the

powerful infrastructure of Google Cloud, we have delivered outstanding results to our customers. Next, we'll leverage its cutting-edge technologies in data analytics and other areas to keep enhancing our services."

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